

100 Days of Azure DevOps

Day 47 Handout — azurerm is Terraform's Azure dialect. features {} is required. Auth today is az login, not a master-key SPN

5 - Infrastructure as Code · Personal lab only · Educational content · Not a sales pitch

Architecture A — provider, auth, resource graph

Piece	HCL / CLI	Maps to
Provider	required_providers hashicorp/azurerm + features {}	ARM APIs. Empty features {} is still required
Auth (lab)	az login; az account set --subscription	Azure CLI token. Terraform uses that context
RG	azurerm_resource_group.lab	Microsoft.Resources/resourceGroups
Storage	azurerm_storage_account.lab account_replication_type LRS	Microsoft.Storage/storageAccounts
Graph	storage.resource_group_name = azurerm_resource_group.lab.name	Wrong ref → plan creates a second RG

Architecture B — identity for apply

Identity	Lab?	Later
Your user via az login	Yes — personal sub, short life	Not how prod pipelines apply
SPN with Owner on subscription	No	Flamethrower. Scope to an RG, prefer WIF (Phase 7)
Service connection in ADO	Day 49	Same least-privilege rule
Pin provider version	Yes (~> 3.x in required_providers)	Floating latest is a surprise engine

Architecture C — pitfalls in the dialect

Pitfall	What you see	Fix
Missing features {}	Provider configuration invalid	provider "azurerm" { features {} }
Wrong subscription	Resources in a work tenant	az account show before apply; personal only
Globally unique name	StorageAccountAlreadyExists	unique suffix; do not reuse a blog's stname
Plan shows extra RG	reference used a string, not the resource	Use .name / .id from the RG resource

One-liner to remember

azurerm is Terraform's Azure dialect — same ideas as ARM/Bicep, different accent. Do not give the dialect a master key to practice grammar.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal az login. Personal subscription. Destroy before you sleep. No work SPN.

- 1 az login and az account show. Confirm the personal subscription ID. az account set if it pointed at the wrong one.
- 2 Extend Day 46: add azurerm_storage_account in rg-day47-tf (new RG name). Pin provider ~> 3.100.
- 3 terraform plan. Confirm one RG + one storage. If you see two RGs, fix the reference.
- 4 terraform apply. az storage account show -g rg-day47-tf -n <name>.
- 5 No subscription-Owner service principal. Your user is enough for a lab.
- 6 terraform destroy -auto-approve. Confirm both resources gone.

azurerm — RG + storage, CLI auth

```
provider "azurerm" {  
  features {}  
}  
  
resource "azurerm_resource_group" "lab" {  
  name      = "rg-day47-tf"  
  location = "Central India"  
}  
  
resource "azurerm_storage_account" "lab" {  
  name                        = "stday47xxxxx" # 3-24 lowercase; change xxxxx so it is globally unique  
  resource_group_name       = azurerm_resource_group.lab.name  
  location                   = azurerm_resource_group.lab.location  
  account_tier               = "Standard"  
  account_replication_type   = "LRS"  
  min_tls_version            = "TLS1_2"  
}  
  
# az login  
# terraform init && terraform plan && terraform apply  
# terraform destroy -auto-approve
```

Done checklist

- az account show was the personal subscription
- Plan showed one RG + one storage via resource references
- features {} present; provider version pinned
- destroy completed

Tomorrow — Day 48

Terraform modules and remote state — backend azurerm, blob lease locking, destroy order.